

Wazuh SIEM Setup Lab

By Michael Ambeguia

Purpose: The purpose of this lab is to set up the open source SIEM/XDR platform Wazuh. I will set up the main components of Wazuh which are the server and agents in this lab. For this lab I will use a pre-built virtual appliance provided by Wazuh for free.

Sections:

1. Intro to Wazuh
2. Set up Wazuh Virtual Appliance
3. Set up a Linux Agent
4. Set up a Windows Agent

Section #1 Intro to Wazuh

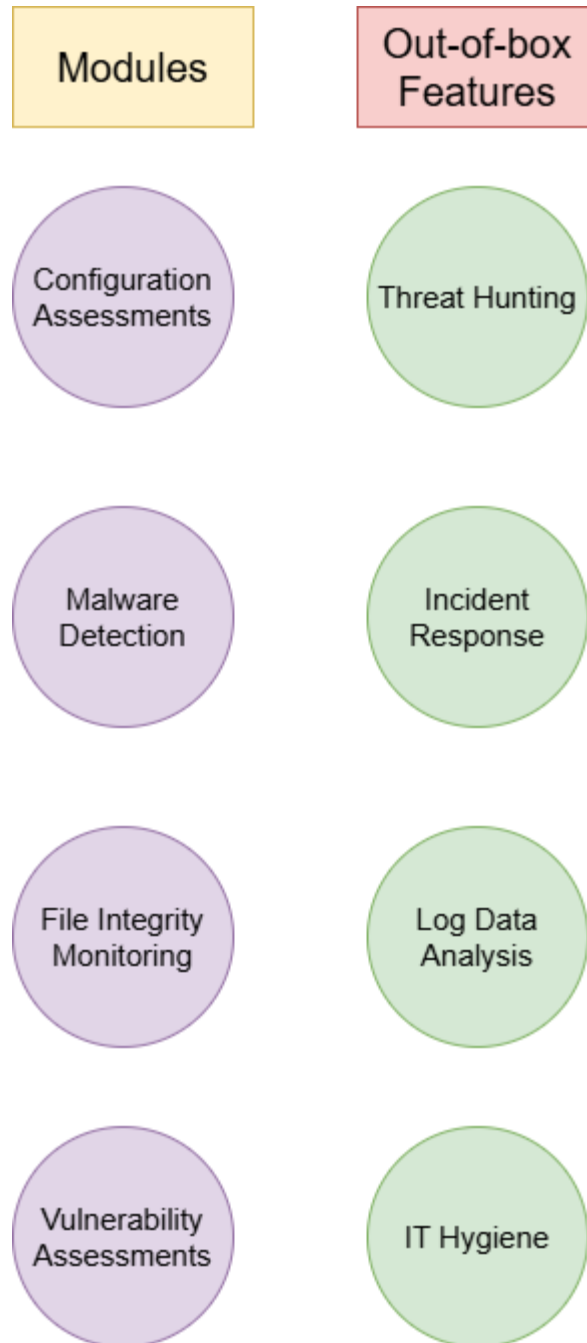
1.1 What is Wazuh?

Wazuh is an open source XDR and SIEM platform that can be used to help organizations monitor their endpoint devices in a centralized manner.

1.2. How is Wazuh open source?

Wazuh is open source since the source code for its platform is free to access and use. You can access the code at Wazuh's GitHub repository at this link: <https://github.com/wazuh/wazuh>. You can also access Wazuh from its website as well.

1.3 What are the features of Wazuh?



1.4 What are the parts of Wazuh's architecture?

1. Wazuh Server: The Wazuh server is responsible for controlling the agents.
 - The Wazuh server allows you to configure the agents remotely.
2. Wazuh Agent: Wazuh Agents are located on endpoints and collect log data and other information from them.

3. Wazuh Indexer: The Wazuh Indexer is used to index and store the alerts created by the Wazuh server.
4. Wazuh Dashboard: The Wazuh dashboard is where the magic happens. The dashboard is where all the data is aggregated and correlated. It is also where the visualization and configuration occurs.

Section #2 Set up Wazuh Appliance:

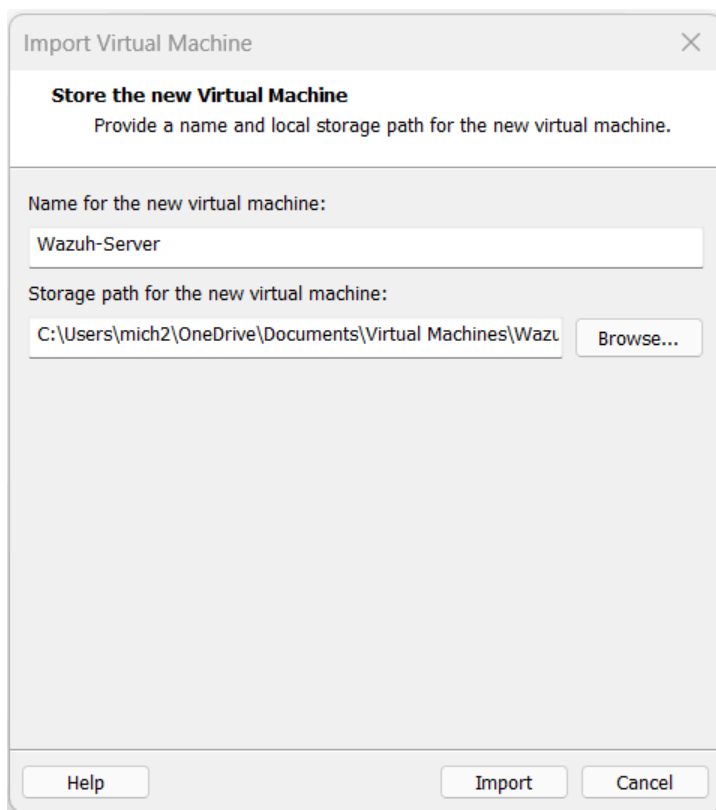
1.1 Download Wazuh appliance ISO:

The Wazuh virtual appliance (VM) can be installed from the following webpage:

<https://documentation.wazuh.com/current/deployment-options/virtual-machine/virtual-machine.html>

I am installing the virtual appliance since it is a convenient way to deploy Wazuh.

1.2 Set up the VM in VMware Workstation Pro:





1.3 Change VM settings:

I will change the amount of memory since 8GB is too much for the purposes of the lab.

 **Wazuh-Server**

[▶ Power on this virtual machine](#)
[🔧 Edit virtual machine settings](#)
[🔄 Upgrade this virtual machine](#)

▼ Devices

 Memory	8 GB
 Processors	4
 Hard Disk (IDE)	50 GB
 CD/DVD (IDE)	Using unknown ...
 Floppy	Using drive A:
 Network Adapter	Bridged (Autom...
 Display	Auto detect

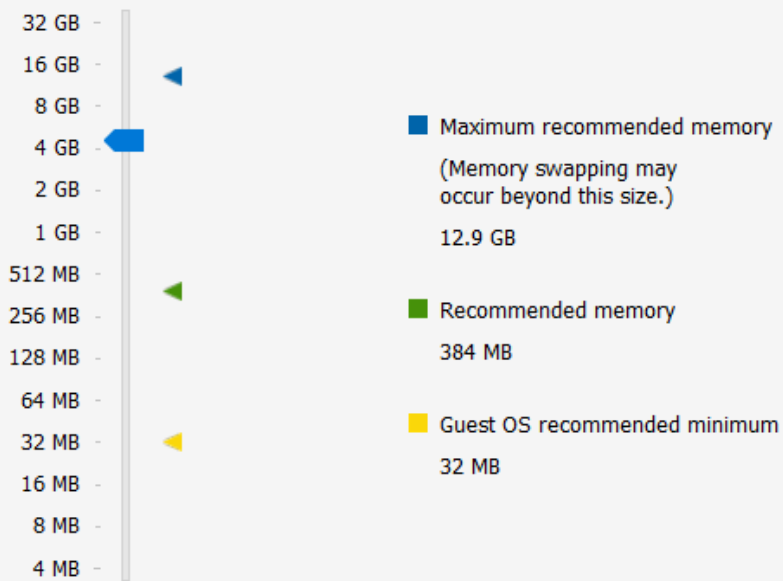
▼ Description

Wazuh enhances security visibility in your infrastructure by monitoring endpoints at the operating system and application levels. Its capabilities include log analysis, file integrity monitoring, intrusion detection, and compliance monitoring.

Memory

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: MB



The image shows a memory allocation interface. On the left is a vertical slider with labels from 4 MB to 32 GB in increments of 4 MB. A blue arrow points to the 4 GB mark. To the right of the slider is a legend with three items: a blue square for 'Maximum recommended memory (Memory swapping may occur beyond this size.) 12.9 GB', a green square for 'Recommended memory 384 MB', and a yellow square for 'Guest OS recommended minimum 32 MB'.

Memory Type	Value
Maximum recommended memory	12.9 GB
Recommended memory	384 MB
Guest OS recommended minimum	32 MB

1.4 Login to the VM:

The appliance already has a user with the following credentials:

```
user: wazuh-user
password: wazuh
```

I logged in as wazuh-user

```
[wazuh-user@wazuh-server ~]$ whoami
wazuh-user
[wazuh-user@wazuh-server ~]$
```

Change password for the wazuh-user on the server VM:

I am able to do this since the wazuh-user has sudo privileges! This user is in the wheel group!

```
[wazuh-user@wazuh-server ~]$ id
uid=1000(wazuh-user) gid=1000(wazuh-user) groups=1000(wazuh-user),10(wheel)
[wazuh-user@wazuh-server ~]$
```

```
[wazuh-user@wazuh-server ~]$ sudo passwd wazuh-user
Changing password for user wazuh-user.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[wazuh-user@wazuh-server ~]$ █
```

1.5 Give the VM a static IP Address:

```
[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:17:6f:81 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.33/24 brd 192.168.1.255 scope global dynamic eth0
        valid_lft 85994sec preferred_lft 85994sec
    inet6 fe80::20c:29ff:fe17:6f81/64 scope link
        valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$ █
```

The Wazuh appliance is an Amazon Linux vm so the network configuration files are located in /etc/sysconfig/network-scripts/. I will have to configure the file with the name of my ethernet interface (eth0).

```
GNU nano 2.9.8 vmimport.ifcfg-eth0
# Created by cloud-init on instance boot automatically, do not edit.
#
BOOTPROTO=dhcp
DEVICE=eth0
HWADDR=08:00:27:7a:b3:d8
NM_CONTROLLED=no
ONBOOT=yes
STARTMODE=auto
TYPE=Ethernet
USERCTL=no
```

I don't want cloud-init so I will disable it then edit this config file. Online I learned that creating the following file will disable it.

```
[wazuh-user@wazuh-server ~]$ touch /etc/cloud/cloud-init.disabled
```

I rebooted the system. Now I will give the server a static ip address.

```
GNU nano 2.9.8 /etc/systemd/scripts/vmimport.ifcfg-eth0
# Created by cloud-init on instance boot automatically, do not edit.
#
BOOTPROTO=none
DEVICE=eth0
IPADDR=192.168.1.33
NETMASK=255.255.255.0
GATEWAY=192.168.1.1
DNS1=1.1.1.1
DNS2=8.8.8.8
HWADDR=08:00:27:7a:b3:d8
NM_CONTROLLED=no
ONBOOT=yes
STARTMODE=auto
TYPE=Ethernet
USERCTL=no
```

1.6 Log into the VM's web portal:

The portal can be accessed via <http://ip-of-the-wazuh-server>

wazuh.

The Open Source Security Platform



Log in

The default credentials are admin/admin.

I am in!

 **W.** Overview

AGENTS SUMMARY

This instance has no agents registered.
Please deploy agents to begin monitoring your endpoints.

⊕ Deploy new agent

LAST 24 HOURS ALERT

Critical severity

0

Rule level 15 or higher

High severity

0

Rule level 12 to 14

Medium severity

0

Rule level 10 to 11

ENDPOINT SECURITY

 **Configuration Assessment**
Scan your assets as part of a configuration assessment audit.

 **Malware Detection**
Check indicators of compromise triggered by malware infections or cyberattacks.

 **Threat Hunting**
Browse through your security alerts, identifying issues and threats in your environment.

1.7 Change the default password for the portal:

Apparently you cannot change the admin password via the Wazuh portal. You have to change the Admin password via the command line on the Wazuh server via a bash script. I had to download it from this page:

<https://documentation.wazuh.com/current/user-manual/user-administration/password-management.html>

```
[wazuh-user@wazuh-server ~]$ ls -l
total 44
-rw-rw-r-- 1 wazuh-user wazuh-user 44203 Feb  8 22:32 wazuh-passwords-tool.sh
[wazuh-user@wazuh-server ~]$
```

```
[wazuh-user@wazuh-server ~]$ chmod 764 wazuh-passwords-tool.sh
[wazuh-user@wazuh-server ~]$ ls -l
total 44
-rwxrw-r-- 1 wazuh-user wazuh-user 44203 Feb  8 22:32 wazuh-passwords-tool.sh
[wazuh-user@wazuh-server ~]$
```

```
[wazuh-user@wazuh-server ~]$ sudo bash wazuh-passwords-tool.sh -u admin -p
08/02/2025 22:44:18 INFO: Updating the internal users.
08/02/2025 22:44:24 INFO: A backup of the internal users has been saved in the /
etc/wazuh-indexer/internalusers-backup folder.
08/02/2025 22:44:24 INFO: Generating password hash
08/02/2025 22:44:35 INFO: The filebeat.yml file has been updated to use the File
beat Keystore username and password.
```

Section #3 Set up Linux Agent:

3.1 Install the Wazuh Agent package onto the Linux device:

On the Wazuh dashboard you can easily deploy agent software onto your devices. Wazuh walks you through the process and it is easy.

Deploy new agent



Select the package to download and install on your system:

 **LINUX**

☐ RPM amd64

☐ RPM aarch64

☐ DEB amd64

☒ DEB aarch64

Because I am going to install the agent software on a Ubuntu VM I need to choose the DEB aarch64 package. Next I need to provide the ip of the Wazuh server and a name for the device.



Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address [?](#)

192.168.1.33

☐ Remember server address



Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name: [?](#)

Ubuntu_Desktop

[?](#) The agent name must be unique. It can't be changed once the agent has been enrolled. [?](#)

Then I need to apply the following command on the device:

4

Run the following commands to download and install the agent:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.10.1-1_arm64.deb && sudo WAZUH_MANAGER='192.168.1.33' WAZUH_AGENT_NAME='Ubuntu-Desktop' dpkg -i ./wazuh-agent_4.10.1-1_arm64.deb
```

I ran the command on the Ubuntu VM:

```
spy2@Laptop:~$ sudo wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.10.1-1_arm64.deb && sudo WAZUH_MANAGER='192.168.1.33' WAZUH_AGENT_NAME='Ubuntu-Desktop' dpkg -i ./wazuh-agent_4.10.1-1_arm64.deb
--2025-02-08 15:39:15-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.10.1-1_arm64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 3.169.231.41, 3.169.231.98, 3.169.231.78, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|3.169.231.41|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11068842 (11M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.10.1-1_arm64.deb'

wazuh-agent_4.10.1- 100%[=====>] 10.56M 201KB/s in 45s

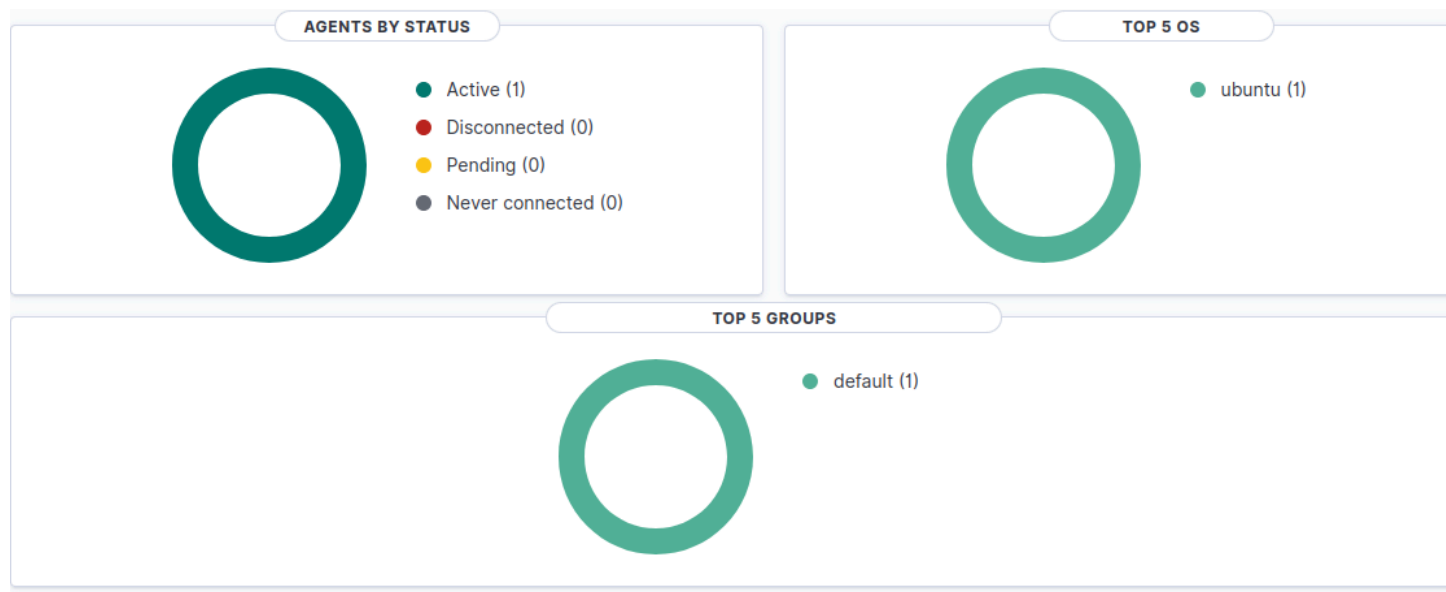
2025-02-08 15:40:01 (241 KB/s) - 'wazuh-agent_4.10.1-1_arm64.deb' saved [11068842/11068842]

Selecting previously unselected package wazuh-agent.
(Reading database ... 206488 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.10.1-1_arm64.deb ...
Unpacking wazuh-agent (4.10.1-1) ...
Setting up wazuh-agent (4.10.1-1) ...
spy2@Laptop:~$
```

Enable the agent so that it starts

```
spy2@Laptop:~$ sudo systemctl daemon-reload
spy2@Laptop:~$ sudo systemctl enable wazuh-agent.service
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /lib/systemd/system/wazuh-agent.service.
spy2@Laptop:~$ sudo systemctl start wazuh-agent.service
spy2@Laptop:~$
```

3.3 Verify that the agent is connected to the server:



Section #4 Set up Windows Agent:

4.1 Install the Wazuh Agent package onto the Windows device:

You can follow the helpful walkthrough on the Wazuh dashboard like before.

Deploy new agent

✓ Select the package to download and install on your system:

LINUX	WINDOWS
☐ RPM amd64	☒ MSI 32/64 bits
☐ DEB amd64	
☐ RPM aarch64	
☐ DEB aarch64	

🔗 For additional systems and architectures, please check our [documentation](#).

You just have to include the server ip and a name for the agent:



Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified c

Assign a server address ?

192.168.1.33



Remember server address



Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent

Assign an agent name: ?

Windows_PC

ⓘ The agent name must be unique. It can't be changed once the agent has been enrolled. [↗](#)

Then you use the command created by Wazuh. Run it on the target machine.

```
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.10.1-1.msi -OutFile $env:tmp\wazuh-agent; msixec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='192.168.1.33' WAZUH_AGENT_NAME='Windows_PC'
```

4.2 Connect the Agent to the Wazuh Server:

```
PS C:\Users\Administrator> NET START WazuhSvc
The Wazuh service is starting.
The Wazuh service was started successfully.

PS C:\Users\Administrator> |
```

4.3 Verify agent is connected to the server:

Agents (1) ⊗ Show only outdated

⊕ Deploy new agent ↻ Refresh ⬇ Export formatted More ▾

WQL

☐	ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
☐	002	Windows_PC	192.168.1.26	default	Microsoft Windows 11 Education 10.0.26100.1742	node01	v4.10.1	● <a>ⓘ <a>👁 <a>⋮	

Usermode Font Driver Host

Wazuh Agent (32 bit)

Windows Defender SmartScreen

0%	1.0 MB	0 MB/s	0 Mbps
0.3%	10.9 MB	0.1 MB/s	0 Mbps
0%	1.3 MB	0 MB/s	0 Mbps